Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	15
interior lacuna count	12
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2753.58	3100.5	1277.32	px^2
major_axis_length_px	lacuna shape	113.949	125.027	36.3344	px
minor_axis_length_px	lacuna shape	33.5852	33.3328	8.5152	px
aspect_ratio	lacuna shape	3.5038	3.3377	1.2462	unitless
eccentricity	lacuna shape	0.9377	0.9529	0.0506	unitless
solidity	lacuna shape	0.7478	0.7578	0.1342	unitless
roots_count	headline	16.3333	15.5	6.6104	count
ring_length_r30_px	headline	519.25	460.5	180.647	px
ring_length_r60_px	reported	1147.08	1087.5	384.549	px
owned_length_px	ownership-dependent	2675.32	2329.52	1165.92	px
edge_count	ownership-dependent	120.917	108	57.617	count
mean_edge_length_px	ownership-dependent	22.67	22.6537	2.6729	px
perimeter_px	normalised	317.48	315.782	100.663	px
ring_area_r30_px2	normalised	11272.2	11500	2573.67	px^2
ring_area_r60_px2	normalised	26792.1	27728	5096.41	px^2
in_frame_fraction_r30	normalised	1	1	0.0001	unitless
in_frame_fraction_r60	normalised	0.9822	1	0.0417	unitless
ring_density_r30	normalised	0.0454785	0.0455332	0.00747507	px^-1
ring_density_r60	normalised	0.0420737	0.0410997	0.00751369	px^-1
roots_per_100px_perimeter	normalised	5.2409	5.2173	1.4801	per 100 px
ring_attached_length_r30_px	network v2	480.417	409.5	180.121	px
ring_attached_length_r60_px	network v2	1007.17	848	413.92	px
ring_length_w_r30_px	network v2	591.28	526.017	209.332	px
ring_length_w_r60_px	network v2	1302.17	1234.24	441.209	px
sholl_crossings_r10	network v2	11.8333	11	4.8021	count
sholl_crossings_r20	network v2	14.0833	13.5	5.2477	count
sholl_crossings_r30	network v2	14.75	14	5.0834	count
loop_count	loops, report only	10	7.5	8.0227	count
loop_edge_share	loops, report only	0.2788	0.2737	0.1201	share
ring_width_mean_r30_px	width	7.5342	7.5749	0.4419	px

Per field
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measure	value	units
analysed_area_px2	1.00906e+06	px^2
total_skeleton_length_px	40081	px
canalicular_length_density_per_px	0.0397211	px^-1
junction_count	917	count
junction_density_per_px2	0.00090877	px^-2
lacuna_count	15	count

lacunae_per_px2	1.487e-05	px^-2
skeleton_component_count	225	count
mean_component_length_px	178.138	px
median_component_length_px	25	px
field_length_density_w_per_px	0.0450005	px^-1
field_density_without_flagged_per_px	0.0396223	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	342	px
unattached_component_count	18	count
canalicular_length_density_attached_per_px	0.0397211	px^-1
field_length_density_with_unattached_per_px	0.04006	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	156	count
loop_edge_share	0.3303	share
edge_count	1971	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2174	px
width_median_px	8.2462	px
width_p10_px	5.6569	px
width_p90_px	10.7703	px
width_pixels_used	38748	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	1001	62.1533	28.1242	2.21	0.8918	0.5533	-0.1968	55.7832	107.949	False
3	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
4	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
5	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
6	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
7	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
8	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
9	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
10	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
11	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
12	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
13	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
14	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
15	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	218	587	843.269	34	24.802	7.8744
2	14	376	696	1457.74	53	27.5045	7.6517

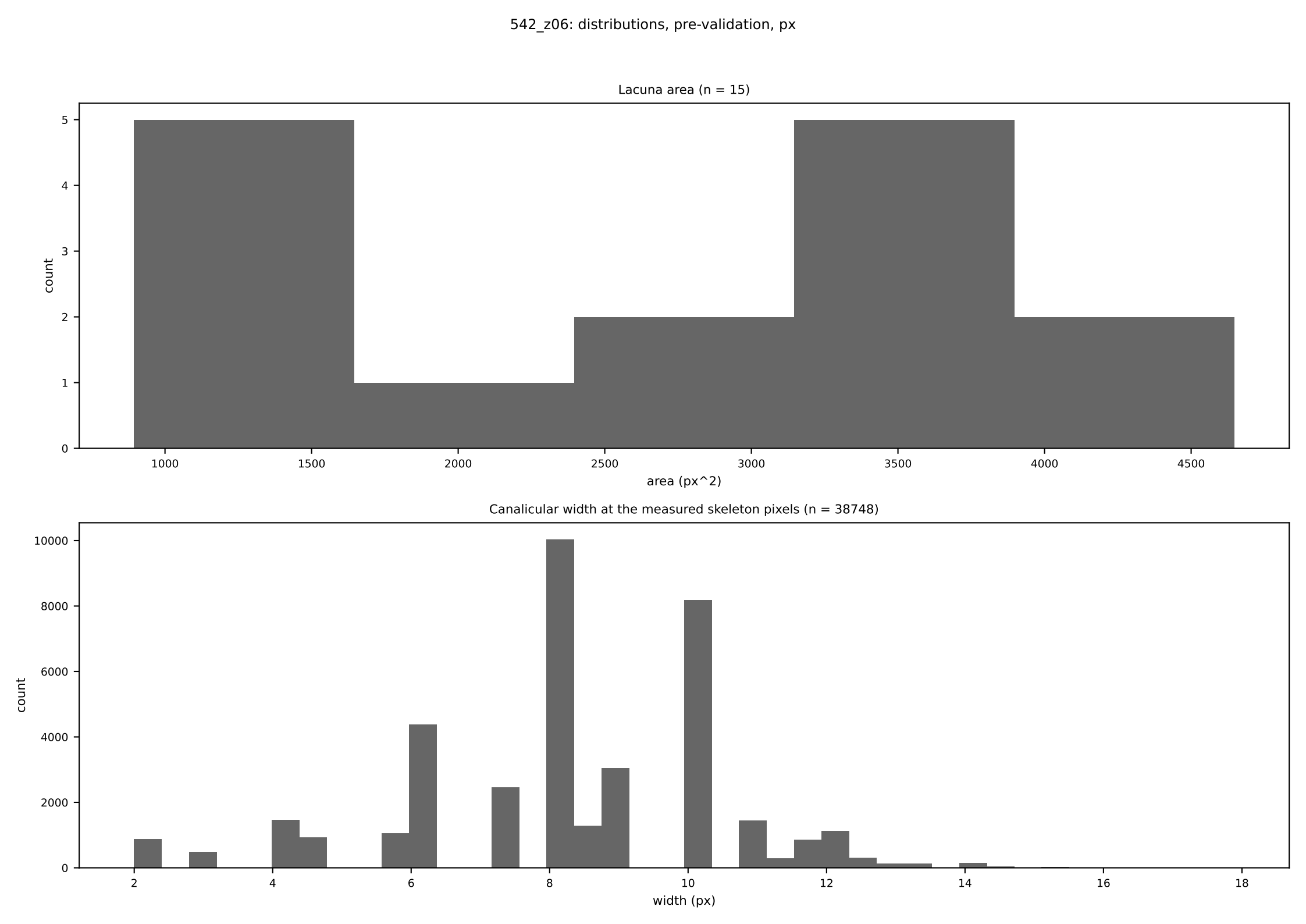
3	11	325	808	1790.02	72	24.8614	7.0688
4	10	462	1118	3647.72	163	22.3787	7.7589
5	29	830	1707	4818.15	205	23.5032	7.6195
6	15	437	890	1199.41	55	21.8074	7.4336
7	17	459	886	1793.42	83	21.6075	7.5303
8	7	420	1057	2338.72	102	22.9287	8.6245
9	16	514	1313	3895.02	160	24.3439	7.62
10	24	870	1902	4169.76	231	18.0509	7.3977
11	23	613	1351	2381.13	114	20.8871	7.0375
12	20	608	1318	2292.4	121	18.9454	7.7263
13	10	317	719	2320.32	92	25.2209	6.9414
14	8	378	910	2468.41	95	25.9833	7.6393
15	5	192	542	200.167	9	22.2407	6.5763

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0388107	0.0462423	3.2523
2	225.309	8379	19327	0.9998	0.884	0.0448741	0.0360118	6.2137
3	151.782	6809	18568	1	1	0.0477309	0.0435157	7.2472
4	322.043	11663	28685	1	1	0.0396125	0.0389751	3.1052
5	438.517	13901	31616	1	1	0.0597079	0.0539917	6.6132
6	211.924	8845	21628	1	1	0.0494064	0.0411504	7.078
7	359.191	12383	27034	1	0.9025	0.037067	0.0327735	4.7329
8	302.451	11295	27170	1	1	0.0371846	0.0389032	2.3144
9	309.522	11337	28286	1	1	0.0453383	0.0464187	5.1693
10	468.274	15363	34096	1	1	0.0566296	0.0557837	5.1252
11	422.978	13165	30001	1	1	0.0465629	0.0450318	5.4376
12	379.848	13296	32108	1	1	0.045728	0.041049	5.2653
13	217.924	8830	22986	1	1	0.0359003	0.0312799	4.5888
14	332.894	10505	23806	0.8602	0.7919	0.0359829	0.0382257	2.4032
15	205.776	6766	16357	0.7841	0.714	0.0283772	0.0331357	2.4298

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	218	580	251.48	670.115	6	5	5
2	316	641	432.404	791.654	8	7	10
3	321	671	367.25	913.696	6	7	10
4	404	958	537.316	1286.86	9	16	15
5	827	1612	956.264	1962.63	22	20	22
6	376	738	491.09	1005.08	8	13	13
7	415	690	514.718	993.08	11	11	10
8	367	650	471.99	1181.62	12	13	13
9	500	1290	590.316	1506.33	11	14	16
10	801	1825	997.566	2153.17	17	22	25
11	561	1169	692.257	1529.35	16	15	15
12	575	1220	688.671	1491.87	15	22	19
13	302	622	355.522	810.713	7	9	9
14	351	746	427.676	1026.47	12	11	16
15	176	205	217.238	605.073	6	7	7


542_z06: distributions, pre-validation, px
Lacuna area (n = 15)
count
area (px^2)
0
1
2
3
4
5
1000
1500
2000
2500
3000
3500
4000
4500
Canalicular width at the measured skeleton pixels (n = 38748)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18